

KING LIVE

Technical Development Proposal

Screen-by-Screen Blueprint · Technical Architecture · SDKs & Packages · Timeline

Consumer App (Flutter) · Reseller Portal · Agency Portal · Master Admin Portal
Live Streaming · Beauty Camera · PK Battles · Gifts · Frames · Entries · Badges · Ranking · Wallet ·
Camera-Presence Enforcement

1. Overview & Recommended Approach

This proposal covers the full King Live scope — live streaming, PK battles, gifting, frames, entries, badges, the reseller/agency/master-admin structure, and the host camera-presence rule — translated into a concrete, buildable structure: every screen the Flutter app and the three web portals need, the SDKs and packages required, and the database shape behind it.

We recommend finalizing the database schema, wallet ledger design, and admin/reseller permission model before writing UI screens. Retrofitting a real-money ledger or a permissions system onto a half-built app is the most common reason a project like this needs a rebuild.

A note on the hardest technical pieces

- Live streaming with multi-host + PK battles at low latency uses a managed RTC provider (Agora or Zego Cloud) — standard practice even for large live-streaming apps, not something built from scratch.
- Production-quality beauty filters (face slim, big eyes, whitening, skin smoothing) require a licensed beauty/AR SDK, not just Flutter alone. Agora and Zego both offer beauty extensions; SenseTime and Meitu are dedicated alternatives — licensed separately from the RTC SDK.
- Apple App Store and Google Play both require in-app purchase (StoreKit / Play Billing) for unlocking virtual currency used inside the app — see Section 8 for how this is structured correctly.
- The camera-presence rule (host must stay visible, niqab permitted) is a person/motion-detection problem, not a face-recognition problem — see Section 6 for the full technical breakdown.

2. Consumer App — Screens (Flutter, Android + iOS)

2.1 Onboarding & Authentication

Screen	Purpose & Key Components
Splash Screen	Logo animation, checks auth token, routes to Login or Home.
Language Select	Optional first-run language picker (Urdu/English/regional).
Login Screen	Phone number or email login; social login optional (Google/Apple, required by App Store if any social login is offered).
Signup Screen	Phone/email + password or OTP-based registration; assigns unique User ID + username.
OTP Verification	SMS/email OTP input, resend timer, auto-read OTP on Android.
Profile Setup Wizard	Name, profile photo upload, gender, country, date of birth, bio — first-login only.
Permissions Screen	Requests camera, microphone, notification, and (Android 13+) media permissions with rationale.

2.2 Home & Discovery

Screen	Purpose & Key Components
Home Shell (Bottom Nav)	Tabs: Popular/Trending, Following, Country/Nearby, New, PK. Persistent bottom nav + top search icon.
Popular / Trending Tab	Grid of active live rooms ranked by viewer count/activity, thumbnail + host name + viewer count badge.
Following Tab	Live rooms only from hosts the user follows; empty-state prompts to discover hosts.
Country / Nearby Tab	Rooms filtered by user's selected/detected country or GPS proximity.
New Hosts Tab	Recently onboarded hosts/streams to boost discovery for new creators.
PK Tab	Grid of rooms currently in an active PK battle, with live score preview on the thumbnail.
Search Screen	Search users by username/ID and rooms by title; recent searches; trending searches.

2.3 Live Room — Host Side

Screen	Purpose & Key Components
Go Live Setup	Choose room type (Single Host / Multi-Guest / Audio Room), set title, cover thumbnail, category, optional room password.
Live Room — Host View	Camera preview with live beauty filter applied, viewer count, comment stream, gift feed, mic/camera toggle, end-live button.

Screen	Purpose & Key Components
Beauty Camera Panel	Sliders/toggles: Skin Smooth, Face Slim, Big Eyes, Whitening, Makeup presets, Skin Tone; global intensity slider; default preset on first use.
AR Filters & Stickers Panel	Face-tracked stickers/AR effects and background blur/replacement, swipeable filter carousel.
Guest Seat Management	Grid of seats (multi-guest mode): invite, remove, mute individual guest, lock/unlock seat.
Host Moderation Controls	Kick viewer, mute viewer, freeze stream (admin-triggered), manage comments on/off, block from room.
Room Settings Sheet	Edit password, select room theme/wallpaper, toggle comments, share room link.
PK Invite Screen	Search/select opponent host, send PK invite, choose duration (Admin-configurable 3/5/10 min).
Presence Monitor (Host)	Background service — silently tracks host-in-frame state, shows on-screen warning banner if absence exceeds threshold (see Section 6).

2.4 Live Room — Viewer Side & PK

Screen	Purpose & Key Components
Live Room — Viewer View	Full-screen stream, floating comments, gift button, follow button, share, viewer count, exit swipe-down.
Gift Panel / Store Sheet	Tabs: Simple Gifts, Premium Gifts, Combo Gifts; balance display; quick-buy coins link if balance low.
Gift Send Animation Overlay	Full-screen animated gift overlay (Lottie/Rive/video) with sender/receiver names, sound effect.
PK Battle View	Split-screen dual host view, live score bars, countdown timer, win-streak/crown badge, rematch/accept-reject prompts.
PK History Screen	Past PK results, win/loss record, win-streak stats.
Room Leaderboard	In-room "Top Senders" ranking, tap to view sender profile.
Entry Effect Overlay	Full-screen animated entrance (Car/Lion/King/Dragon/Jet Entry) + name/ID banner when a VIP user joins the room.

2.5 Social & Profile

Screen	Purpose & Key Components
My Profile Screen	Avatar, level badges, wealth/VIP level, active frame, follower/following counts, edit button, wallet shortcut.

Screen	Purpose & Key Components
Other User Profile	Same layout as own profile plus Follow/Unfollow, Message, Send Gift, Report/Block actions.
Edit Profile Screen	Edit photo, name, bio, gender, country; save validation.
Followers / Following Lists	Paginated user lists with follow-state buttons.
Level & Badges Screen	User Level, Wealth Level, VIP Level progress bars; owned badge showcase grid (3D badge renders).
Frames Gallery	Owned + purchasable frames, preview-on-avatar, apply/remove, expiry countdown per frame.
Entry Effects Gallery	Owned + purchasable entry effects, preview animation, apply/remove, expiry countdown.
Block / Report Screen	Reason picker, evidence note, submit to moderation queue.
Privacy Settings	Who can message me (Everyone/Followers/Nobody), who can see profile/followers, block list management.

2.6 Private Messaging (PM)

Screen	Purpose & Key Components
Chat Inbox (PM List)	List of conversations, unread badge, seen/unseen indicator, last-message preview.
Chat Conversation	Text, emoji, photo, voice message bubbles; in-chat gift send; typing indicator.
Chat Notification Settings	Mute conversation, notification preview toggle.
Blocked Users List	Manage blocked contacts; unblock action.

2.7 Wallet, Recharge & Store

Screen	Purpose & Key Components
Wallet Home	Coin balance, Diamond/Bean balance (for hosts), quick actions: Recharge, Withdraw, History.
Recharge Screen	Coin packages, price list, payment method selection (JazzCash, Easypaisa, bank, card, platform IAP where required).
Payment Method Screens	Provider-specific flow (JazzCash/Easypaisa redirect or in-app SDK, card entry, Apple/Google IAP sheet).
Withdrawal Request (Host)	Enter amount, choose payout method, view fees/conversion rate, submit request.
Transaction History	Filterable ledger: recharge, gift sent/received, withdrawal, admin adjustments — each entry timestamped.

Screen	Purpose & Key Components
Store Screen	Browse/purchase Frames, Entries, Wallpapers, Badges outside of gifting context.

2.8 Ranking & Leaderboards

Screen	Purpose & Key Components
Leaderboard Home	Tabs: Top Sender, Top Host, Top Agency, Top PK Host — each with Daily/Weekly/Monthly toggle.
Country Ranking	Same leaderboard categories filtered by country.
Rank Reward Screen	Shows Top 1/2/3 rewards (special Frame/Badge/Entry) and countdown to next reset.

2.9 Settings, Support & Agency (Host-side)

Screen	Purpose & Key Components
Notifications Center	Follow, Gift, PM, PK Invite, Host Live, Agency, Salary, Withdrawal notifications feed.
Settings / Account	Change password, linked accounts, delete account, language, theme.
Help & Support	FAQ, contact support/ticket, report a bug.
Terms & Community Guidelines	Static content screens required for app store compliance.
Agency Dashboard (Host)	For hosts under an agency: live hours, live days, coin/bean totals vs. target, salary progress bar.

3. Reseller Portal — Screens (Web)

A separate login surface issued and permission-controlled by the Master Admin. Every action here is scoped by whatever the Admin has explicitly allowed for that reseller.

Screen	Purpose & Key Components
Reseller Login	Separate portal login issued by Master Admin.
Reseller Dashboard	Available balance, credit limit, today's sales snapshot, quick actions.
Coin Recharge / Transfer	Search by User ID, enter amount, confirm — deducts from reseller balance/quota.
Frame Assignment	Search User ID, pick from Admin-allowed frame catalog, set duration (1/7/30 days/custom/permanent), assign.
Entry Assignment	Same flow as Frame Assignment for Entry Effects.
Badge Assignment (if permitted)	Manually apply an allowed badge to a User ID, subject to Admin-granted permission.
Remove Frame/Entry/Badge	Only visible if Admin has granted the reseller removal permission.
Sales Report	Daily/Weekly/Monthly sales totals, filterable by item type.
Commission Report	Reseller's earned commission/profit breakdown.
Transaction History	Full log of recharges, frame/entry/badge assignments with User ID, Item ID, dates, status.
User Search	Look up a User ID to check basic status before assigning anything.
Limits View	Read-only view of Admin-set daily assignment limits.

4. Agency Portal — Screens (Web)

Screen	Purpose & Key Components
Agency Login	Agency Owner portal login.
Agency Dashboard	Overview of hosts, total live hours, monthly commission summary.
Host Management	Add/remove hosts, approve host applications, view host status.
Host Performance	Per-host live hours, live days, coins/beans earned vs. target.
Salary & Commission Report	Monthly salary calculation, commission report, payout status.
Agency Ranking	Agency's position on the Top Agency leaderboard; PK/event participation history.

5. Master Admin Portal — Screens (Web)

Role-restricted (RBAC), 2FA-protected for sensitive roles. This is the control center for every other module in the app.

Screen	Purpose & Key Components
Admin Login (2FA)	Email/username + password + OTP/2FA for sensitive-role accounts.
Admin Dashboard	KPIs: DAU/MAU, revenue, active rooms, pending withdrawals, flagged reports.
User Management	Search, ban/unban, device ban, profile reset, view full activity log.
Host Management	Approve/suspend hosts, set salary/target parameters.
Agency Management	Create/approve agencies, set commission rules, control agency's hosts.
Reseller Management	Create resellers, set balance/credit limit, configure granular permissions (recharge, frame, entry, badge, expiry, daily limits).
Coins/Beans Management	Manual add/remove, exchange-rate configuration, full audit log.
Gifts Management	Upload gift asset (3D/animated), set price/category, enable/disable, mark as event/limited-time.
Frames Management	Upload 3D frame asset, set price, eligibility (role-based), expiry rules, reseller access toggle.
Entries Management	Upload entry animation, set price, eligibility, expiry, reseller access toggle.
Badges Management	Upload 3D badge asset; configure Auto Badge System thresholds; manual assign fallback.
Wallpapers Management	Upload static/animated room themes, category tagging, enable/disable.
PK / Ranking Rules	Configure PK timer options, event PK types, leaderboard reward rules, rank reset schedule.
Withdrawal Management	Review pending payout requests, approve/reject, record payment reference, per-country payout config.
Notifications / Broadcast	Compose push notification or in-app announcement, target all users or by country.
Reports Center	User, revenue, recharge, gift, host, agency, and reseller reports with date-range export.
Roles & Permissions (RBAC)	Define Super Admin / Admin / Finance Admin / Official / Support / Moderator roles and module-level access.
Admin Action Logs	Immutable audit trail: which admin changed what, when, from which device/IP.
Moderation Queue	Review reported users/live rooms/chats, apply mute/kick/temp-ban/permanent-ban/device-ban, configure bad-word filter.
Host Presence Rules	Configure camera-presence grace period (default 30s), warning text/behavior, repeat-offender rule, auto-off toggle (see Section 6).

6. Host Camera-Presence Rule — Technical Structure

Business rule (as specified)

- Host must remain visibly present in front of the camera during live streaming.
- A host wearing niqab is fully permitted — niqab must never trigger a warning or an automatic live/ID shutdown.
- The system only verifies that the host's body is present in the camera frame — full facial visibility is never a requirement.
- Maximum allowed continuous camera-absence: 30 seconds.
- If the host returns within 30 seconds, the live stream continues uninterrupted.
- If absence exceeds 30 seconds, the system issues an automatic on-screen warning.
- If the host still has not returned after the warning, the system automatically ends the live stream / freezes the host ID.
- Repeat-offender handling is admin-configurable.

What the system must detect

- Whether the host's body is present in the camera frame — not whether a face is visible.
- Whether a black screen is being shown instead of a live feed.
- Whether an empty chair or background is being shown in place of the host.
- Whether the host is completely outside the frame.
- Whether presence is being faked using a static photo or image held up to the camera.

This is deliberately a presence/motion problem, not a face-recognition problem — a solution built only around face detection would incorrectly flag a niqab-wearing host as "absent," which directly violates the stated rule. The detection pipeline below is designed around that constraint.

6.1 Detection signals (combined — no single signal is reliable alone)

Signal	What it catches	Approach / Package
Body / person presence	Host physically in frame — works even when fully veiled, since it does not rely on a visible face.	google_mlkit_pose_detection (ML Kit Pose Detection) or a lightweight TFLite person-detector.
Motion / frame variance	Catches a static photo held up to the camera, or a frozen frame.	Frame-diff — compare pixel histograms / luminance variance every few frames. No ML model required.
Black-screen / blank check	Catches a covered lens, blocked camera, or empty background shown in place of the host.	Average brightness + variance sampling on captured frames.

6.2 Where the check should run

Client-side (on the host's device)

Fast and inexpensive to build; the host's own device runs detection and reports present/absent status to the server. Suitable for an MVP. Limitation: a modified client app could always report "present," which defeats the

rule's purpose as a real anti-fraud control.

Server-side (recommended for real enforcement)

The backend periodically pulls a snapshot from the actual outgoing live stream (via Agora Cloud Recording or the Zego snapshot API) and runs detection on that snapshot. The host's client cannot spoof this, since the server is checking the real broadcast stream rather than a self-reported status. This requires an additional backend worker service and is recommended as a Phase 2 hardening step rather than a Day 1 requirement.

6.3 State machine

State	System Behavior
Present	Live continues normally — no action.
Niqab worn, body visible	Live continues normally — this is not a violation.
Absent 0–30 seconds	Internal countdown running — no visible action to the host or viewers.
Absent > 30 seconds	System fires an automatic on-screen warning to the host.
Still absent after warning	System automatically ends the live stream / freezes the host ID; event logged.
Repeat offense	Admin-configurable rule (e.g. escalating penalty, temporary suspension) applied via the Admin panel.

6.4 Admin controls

- Configurable grace period (default 30 seconds).
- Configurable warning message / on-screen behavior.
- Repeat-offender rule configuration (escalation policy).
- Toggle to enable/disable auto-off entirely per host, per agency, or platform-wide.
- Full audit log of every presence violation and system action taken.

6.5 Development effort for this feature

Component	Estimated Effort	
Client-side detection pipeline (pose + motion + brightness)	4–6 days	
State machine + host-side warning UI	2–3 days	
Admin config screen + repeat-offender rule + logging	2–3 days	
Server-side snapshot verification (recommended hardening)	+1–2 weeks	

Recommendation: build the client-side version first for the MVP/demo, and scope server-side verification as an explicit Phase 2 item so both sides are clear on what is and isn't spoof-resistant at launch.

7. Technical Architecture

7.1 Flutter App — Packages & Services

Component	Package / SDK
State management	get (GetX) — controllers, routing, dependency injection.
Live streaming engine	agora_rtc_engine or zego_express_engine — managed SFU for low-latency multi-host + PK streaming.
Beauty / AR filters	Licensed vendor SDK (Agora Extensions Marketplace, Zego Effects, SenseTime, or Meitu) for production-quality Face Slim/Big Eyes/Whitening/Makeup.
Presence / pose detection	google_mlkit_pose_detection (body-keypoint detection, works without a visible face) — powers the camera-presence rule.
Face landmark detection (optional AR)	google_mlkit_face_detection — only if custom stickers beyond the beauty SDK are needed.
Networking	dio + typed API layer, interceptors for auth token refresh.
Realtime chat / presence	socket_io_client (self-hosted backend) or Firebase Realtime Database/Firestore streams.
Local storage / cache	hive or get_storage for session, drafts, lightweight cache.
Push notifications	firebase_messaging.
Animations (gifts/entries/frames)	rive and/or lottie for full-screen animated overlays.
Image loading/caching	cached_network_image.
Payments — local (PK)	Direct REST integration with JazzCash and Easypaisa merchant APIs (server-verified webhooks).
Payments — store compliance	in_app_purchase (StoreKit/Play Billing) — required by Apple/Google for in-app virtual currency.
Analytics & crash reporting	firebase_analytics, firebase_crashlytics (or Sentry).
Image compositing	image (Dart) — used to composite sticker overlays onto captured frames.

7.2 Backend & Infrastructure

Component	Technology
Application backend	Node.js (NestJS) or Laravel — REST/GraphQL API serving Flutter app + web Admin/Reseller/Agency portals.
Primary database	PostgreSQL (or MySQL) — Users, Wallet, Transactions, Gifts, Frames, Entries, Badges, Reseller/Agency data.

Component	Technology
Cache / realtime counters	Redis — session cache, live viewer counts, leaderboard sorted sets, rate limiting.
Chat storage	MongoDB or dedicated chat tables at high PM/comment volume; Postgres sufficient at moderate scale.
Live streaming infrastructure	Agora or Zego Cloud (managed SFU/media servers).
Presence-detection worker	Small backend service pulling periodic snapshots from the RTC stream (Agora Cloud Recording / Zego snapshot API) and running the pose/motion check server-side.
Admin / Reseller / Agency portals	Web app (React or Vue) — separate frontend, same backend API.
Object storage & CDN	AWS S3 / Cloudflare R2 for assets, fronted by CloudFront/Cloudflare.
Push delivery	Firebase Cloud Messaging (Android + iOS).
Payments	JazzCash & Easypaisa merchant APIs (server-verified webhooks), plus Apple/Google IAP reconciliation.
Monitoring & backups	Grafana/Prometheus or hosted APM, automated DB backups, documented rollback strategy.

7.3 Suggested Flutter Project Structure

Clean architecture layering with GetX for state/DI — appropriate for a module count this large:

lib/app/ — routes, GetX bindings per screen

lib/core/ — camera, RTC engine wrapper, beauty SDK bridge, presence-detection service, theme, utils

lib/data/ — models, repositories, remote (API) + local (cache) datasources

lib/domain/ — entities, use-cases (SendGiftUseCase, JoinPKUseCase, CheckHostPresenceUseCase, etc.)

lib/presentation/ — one folder per screen group above, each with its own GetX controller

7.4 Core Database Tables

Wallet balance is never stored as a single mutable number without a transaction ledger; frames/entries/badges are reusable catalogs with separate assignment records so expiry and eligibility are tracked independently.

Table	Purpose
users	profile fields, level, wealth_level, vip_level, country, status flags, device IDs.
wallets	coin_balance, diamond_balance.
transactions	immutable ledger of every credit/debit: type, amount, ref IDs, timestamp.
gifts / gift_events	gift catalog and every send event (sender, receiver, room, amount, timestamp).

Table	Purpose
frames / entries / badges	reusable catalogs — id, asset, eligibility rule, price, active flag.
user_items	assignment records linking user_id to a frame/entry/badge, with expiry, status, source.
rooms / pk_battles	live room state; PK duration, scores, winner, streak.
presence_events	per-host camera-presence log: timestamp, state, action taken, triggering source (client/server).
resellers / agencies	balance, credit limit, permission flags, commission rate, host roster.
withdrawal_requests	host_id, amount, method, status, admin_reviewer_id.
admin_action_logs	admin_id, action, target, before/after, device/IP, timestamp.

8. Store Compliance — Read Before Building Payments

- Apple and Google both classify coins/diamonds used for virtual gifts as "virtual currency" / digital content. In-app coin purchases must route through StoreKit (iOS) and Play Billing (Android) — direct JazzCash/Easypaisa/card checkout inside the app binary is not permitted for this.
- JazzCash/Easypaisa/bank transfer can still power web-based recharge (outside the app binary) and the Reseller Portal's coin-transfer flow, since that is a B2B/portal action rather than an in-app consumer purchase.
- Live one-to-one/group video with virtual gifting receives extra App Store review scrutiny — plan for content moderation, an in-app reporting flow, and block/report at first submission, not as a post-launch patch.
- Host payout/withdrawal (converting diamonds/beans to real money) is generally fine outside IAP, since it is a payout to the creator rather than a consumer purchase — but keep it clearly separated from the coin-purchase path in both UI and code.

9. Development Phases

Phase	Scope
Phase 0 — Foundation	Database schema, wallet/ledger design, RBAC permission model, overall system architecture.
Phase 1 — Core	Auth, Profile, Live Streaming (single host), Beauty Camera, Follow, PM, Coins, Gifts, basic Admin panel.
Phase 2 — Business Layer	PK Battles, Ranking, Agency module, Reseller module, Salary logic, Frames, Entries, Wallpapers, Withdrawals.
Camera-Presence Rule	State machine (present / grace / warning / auto-off), host-side warning UI, admin config screen, logging (client-side first, server-verified as hardening).
Phase 3 — Premium	VIP levels, Badge system, full-screen gift animations, advanced rankings, special events, audio rooms, analytics.
Phase 4 — Scale (post-MVP)	Performance tuning, CDN rollout, auto-scaling, server-side presence verification, advanced moderation, backup/DR drills.

9.1 Estimated Timeline

This estimate covers the full MVP — Phases 0 through 3 plus the camera-presence rule. Phase 4 (scale/hardening) is a post-launch item, scoped and quoted separately once the MVP is live.

Milestone	Estimated Duration	
Phase 0 — Foundation	2–3 weeks	
Phase 1 — Core	6–8 weeks	
Phase 2 — Business Layer	6–8 weeks	
Camera-Presence Rule (standard version)	1.5–2 weeks	
Camera-Presence Rule (advanced, tamper-resistant version)	+1–2 weeks	
Phase 3 — Premium	4–6 weeks	
Estimated MVP Total	~20–29 weeks (5–7 months)	

Factors that can affect this timeline

- Beauty/RTC SDK licensing approval and first-integration work typically add 1–2 weeks not visible in a task-by-task breakdown.
- App Store / Play Store review cycles for live-streaming + virtual-currency apps commonly involve multiple review rounds — we recommend budgeting 3–4 weeks after development is functionally complete for store approval.
- This estimate covers a working MVP; dedicated QA/bug-fix cycles and production-scale hardening (load testing, security audit) are scoped separately under Phase 4.

10. Recommended Build Order

(1) Database schema + wallet ledger + RBAC permission model + overall architecture, (2) Live/PK streaming architecture with the chosen RTC provider, (3) Reseller/Agency permission and assignment logic, (4) UI screens module by module following the phase table above, with the camera-presence rule built alongside the Live Room screens in Phase 1–2. Building UI before the ledger and permission model are locked is the most common reason a project like this needs a second rebuild.